

DBE&BE PRACTICAL – 3

2520090046

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1. Document Modelling:

```
>_MONGOOSH
  formats: ["Hardcover", "Digital", "Audiobook"],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: "K. Rao",
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: "M1001",
      rating: 5,
      comment: "Excellent deep dive into digital systems design.",
      date: ISODate("2023-01-15")
    },
    {
      member_id: "M1042",
      rating: 4,
      comment: "Good but the Digital edition had formatting issues.",
      date: ISODate("2023-02-20")
    }
  ]
});
< {
  acknowledged: true,
  insertedId: ObjectId('6a8d20a1e9b1256610eda46f')
}
```

```
>_MONGOOSH
  avgRating: 3.8,
  reviews: [
    { member_id: "M3001", rating: 3, comment: "Slow start but strong ending.", date: ISODate("2020-01-15") },
    { member_id: "M3002", rating: 4, comment: "Beautifully written prose.", date: ISODate("2020-02-11") }
  ],
  {
    title: "Quantum Horizons",
    author: "R. Menon",
    genre: "Science",
    publishedYear: 2023,
    formats: ["Digital", "Audiobook"],
    specs: { fileSize_MB: 5.6, drm: true, narrator: "P. Iyer", runtime_min: 540 },
    avgRating: 4.9,
    reviews: [
      { member_id: "M4001", rating: 5, comment: "The Digital format made annotations so easy.", date: ISODate("2023-08-01") },
      { member_id: "M4002", rating: 5, comment: "Mind-bending and well narrated.", date: ISODate("2023-09-14") }
    ]
  }
});
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8d217ee9b1256610eda470'),
    '1': ObjectId('6a8d217ee9b1256610eda471'),
    '2': ObjectId('6a8d217ee9b1256610eda472')
  }
}
```

2. Advanced Querying

a) Books rated above 4 (\$gt)

```
>_MONGOSH
    { member_id: "M4002", rating: 5, comment: "Mind-bending and well narrated.",
      }
    }
  });
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a8d217ee9b1256610eda470'),
    '1': ObjectId('6a8d217ee9b1256610eda471'),
    '2': ObjectId('6a8d217ee9b1256610eda472')
  }
}
> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
);
< {
  title: 'The Silent Algorithm',
  avgRating: 4.6
}
{
  title: 'Ocean of Code',
  avgRating: 4.2
}
{
  title: 'Quantum Horizons',
  avgRating: 4.9
}
}
```

b) Books available in specific formats (\$in):

```
>_MONGOSH
  avgRating: 4.9
}
> db.book_metadata.find(
  { formats: { $in: ["Digital", "Audiobook"] } },
  { title: 1, formats: 1, _id: 0 }
);
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ]
}
{
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ]
}
}
```

c) Combine both with \$and (implicit):

```
>_MONGOSH
}
> db.book_metadata.find({
  avgRating: { $gt: 4 },
  formats: { $in: ["Digital"] }
});
< [
  {
    _id: ObjectId('6a8d20a1e9b1256610eda46f'),
    title: 'The Silent Algorithm',
    author: 'R. Menon',
    genre: 'Technology',
    publishedYear: 2022,
    formats: [
      'Hardcover',
      'Digital',
      'Audiobook'
    ],
    specs: {
      pages: 340,
      fileSize_MB: 4.2,
      drm: true,
      narrator: 'K. Rao',
      runtime_min: 610
    },
    avgRating: 4.6,
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
```

```
>_MONGOSH
genre: 'Science',
publishedYear: 2023,
formats: [
  'Digital',
  'Audiobook'
],
specs: {
  fileSize_MB: 5.6,
  drm: true,
  narrator: 'P. Iyer',
  runtime_min: 540
},
avgRating: 4.9,
reviews: [
  {
    member_id: 'M4001',
    rating: 5,
    comment: 'The Digital format made annotations so easy.',
    date: 2023-08-01T00:00:00.000Z
  },
  {
    member_id: 'M4002',
    rating: 5,
    comment: 'Mind-bending and well narrated.',
    date: 2023-09-14T00:00:00.000Z
  }
]
]
```

d) Regex query — books with "Digital" format:

```
>_MONGOSH
> db.book_metadata.find(
  { formats: { $regex: "Digital", $options: "i" } },
  { title: 1, formats: 1 }
);
< {
  _id: ObjectId('6a8d20a1e9b1256610eda46f'),
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ]
}
{
  _id: ObjectId('6a8d217ee9b1256610eda470'),
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ]
}
{
  _id: ObjectId('6a8d217ee9b1256610eda472'),
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ]
}
```

e) Regex query — reviews mentioning "Digital" in comments (nested field):

```

>_MONGOSH
> db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } },
  { title: 1, "reviews.$": 1 }
);
< [
  {
    _id: ObjectId('6a8d20a1e9b1256610eda46f'),
    title: 'The Silent Algorithm',
    reviews: [
      {
        member_id: 'M1001',
        rating: 5,
        comment: 'Excellent deep dive into digital systems design.',
        date: 2023-01-15T00:00:00.000Z
      }
    ]
  },
  {
    _id: ObjectId('6a8d217ee9b1256610eda470'),
    title: 'Ocean of Code',
    reviews: [
      {
        member_id: 'M2001',
        rating: 4,
        comment: 'Solid Digital read, clean layout.',
        date: 2022-05-10T00:00:00.000Z
      }
    ]
  }
]

```

```

>_MONGOSH
    date: 2023-01-15T00:00:00.000Z
  }
]
}
{
  _id: ObjectId('6a8d217ee9b1256610eda470'),
  title: 'Ocean of Code',
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: 2022-05-10T00:00:00.000Z
    }
  ]
}
}
{
  _id: ObjectId('6a8d217ee9b1256610eda472'),
  title: 'Quantum Horizons',
  reviews: [
    {
      member_id: 'M4001',
      rating: 5,
      comment: 'The Digital format made annotations so easy.',
      date: 2023-08-01T00:00:00.000Z
    }
  ]
}
]

```

Task 3: Aggregation Pipeline — "Top Rated" Report

```

>_MONGOSH
> db.book_metadata.aggregate([

  { $match: { publishedYear: { $gt: 2020 } } },

  { $unwind: "$reviews" },

  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  },

  { $sort: { avgReviewRating: -1 } },

  {
    $project: {
      _id: 0,
      title: "$_id",
      author: 1,
      avgReviewRating: { $round: ["$avgReviewRating", 2] },
      totalReviews: 1
    }
  }
])

```

```

>_MONGOSH
{
  $project: {
    _id: 0,
    title: "$_id",
    author: 1,
    avgReviewRating: { $round: ["$avgReviewRating", 2] },
    totalReviews: 1
  }
}
]);
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
}
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}
}

```

Task 4: Indexing for Speed

- a) Create a text index on review comments

```

}
> db.book_metadata.createIndex({ "reviews.comment": "text" });
< reviews.comment_text

```

- b) Query using the text index:

```

>_MONGOSH
}
[
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
]
> db.book_metadata.createIndex({ "reviews.comment": "text" });
< reviews.comment_text
> db.book_metadata.find(
  { $text: { $search: "Digital" } },
  { title: 1, score: { $meta: "textScore" } }
).sort({ score: { $meta: "textScore" } });
< [
  _id: ObjectId('6a8d20a1e9b1256610eda46f'),
  title: 'The Silent Algorithm',
  score: 1.1833333333333333
]
[
  _id: ObjectId('6a8d217ee9b1256610eda472'),
  title: 'Quantum Horizons',
  score: 0.6
]
[
  _id: ObjectId('6a8d217ee9b1256610eda470'),
  title: 'Ocean of Code',
  score: 0.6
]

```

c) Verify index usage with .explain()

```

>_MONGOSH
> db.book_metadata.find(
  { $text: { $search: "Digital" } }
).explain("executionStats");
< [
  explainVersion: '1',
  queryPlanner: {
    namespace: '2520080053_Ruhitha.book_metadata',
    parsedQuery: {
      '$text': [
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      ]
    },
    indexFilterSet: false,
    queryHash: '4FB85230',
    planCacheShapeHash: '4FB85230',
    planCacheKey: 'F34F3870',
    optimizationTimeMillis: 1,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      ns: '2520080053_Ruhitha.book_metadata',
      indexPrefix: {}
    }
  }
]

```

```

>_MONGOSH
  indexName: 'reviews.comment_text',
  parsedTextQuery: {
    terms: [
      'digit'
    ],
    negatedTerms: [],
    phrases: [],
    negatedPhrases: []
  },
  textIndexVersion: 3,
  inputStage: {
    stage: 'FETCH',
    nss: '2520080053_Ruhitha.book_metadata',
    inputStage: {
      stage: 'IXSCAN',
      nss: '2520080053_Ruhitha.book_metadata',
      keyPattern: {
        _fts: 'text',
        _ftsx: 1
      },
      indexName: 'reviews.comment_text',
      isMultiKey: true,
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'backward',
      indexBounds: {}
    },
    indexName: 'reviews.comment_text',
    isMultiKey: true,
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {}
  },
  indexName: 'reviews.comment_text',
  isMultiKey: true,
  isUnique: false,
  isSparse: false,
  isPartial: false,
  indexVersion: 2,
  direction: 'backward',
  indexBounds: {}
}

```

```

>_MONGOSH
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 6,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'TEXT_MATCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: '2520080053_Ruhitha.book_metadata',
    indexPrefix: {},
    indexName: 'reviews.comment_text',
    parsedTextQuery: {
      terms: [
        'digit'
      ],
      negatedTerms: [],
      phrases: [],
      negatedPhrases: []
    }
  }
}

```

```

>_MONGOSH
    negatedPhrases: []
  },
  textIndexVersion: 3,
  docsRejected: 0,
  inputStage: {
    stage: 'FETCH',
    nReturned: 3,
    executionTimeMillisEstimate: 0,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: '2520080053_Ruhitha.book_metadata',
    docsExamined: 3,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 4,
      advanced: 3,
      needTime: 0,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,

```

```

>_MONGOSH
    nss: '2520080053_Ruhitha.book_metadata',
    keyPattern: {
      _fts: 'text',
      _ftsx: 1
    },
    indexName: 'reviews.comment_text',
    isMultiKey: true,
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'backward',
    indexBounds: {},
    keysExamined: 3,
    seeks: 1,
    dupsTested: 3,
    dupsDropped: 0,
    peakTrackedMemBytes: 26
  }
}
),
queryShapeHash: 'F3AD0A3A4DE07E9DFF04007ABEE44B1166B74E6B4280F46E28830220186D407',
peakTrackedMemBytes: Long('26'),
command: {
  find: 'book_metadata',
  filter: {
    '$text': {
      '$search': 'Digital'

```

```

>_MONGOSH
find: 'book_metadata',
filter: {
  $text: {
    $search: 'Digital'
  }
},
'$db': '2520080053_Ruhitha'
},
serverInfo: {
  host: 'Ruhitha',
  port: 27017,
  version: '3.3.9',
  gitVersion: '35e8c8a57f7815fed9fac1a9e9e6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnHint: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFromSourceControl: 'querySourceRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}

```

d) Compare against a non-indexed regex scan

```

>_MONGOSH
> db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } }
).explain("executionStats");
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: '2520080053_Ruhitha.book_metadata',
    parsedQuery: {
      'reviews.comment': BSONRegExp('Digital', 'i')
    },
    indexFilterSet: false,
    queryHash: '819E6067',
    planCacheShapeHash: '819E6067',
    planCacheKey: '29FFD58A',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: {
        'reviews.comment': BSONRegExp('Digital', 'i')
      },
      nss: '2520080053_Ruhitha.book_metadata',
      direction: 'forward'
    },
    rejectedPlans: []
  }
}

```

```

>_MONGOSH
  executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 1,
    totalKeysExamined: 0,
    totalDocsExamined: 4,
    executionStages: {
      isCached: false,
      stage: 'COLLSCAN',
      filter: {
        'reviews.comment': BSONRegExp('Digital', 'i')
      },
      nReturned: 3,
      executionTimeMillisEstimate: 0,
      works: 5,
      advanced: 3,
      needTime: 1,
      needYield: 0,
      saveState: 0,
      restoreState: 0,
      isEOF: 1,
      nss: '2520080053_Ruhitha.book_metadata',
      direction: 'forward',
      docsExamined: 4
    }
  },
  queryShapeHash: 'DE58F0EA1F9D0F0BEC392841E964F3E0A1D246F6636711090F560508D0DE2625',
  command: {
    find: 'book_metadata'
  }

```

```

>_MONGOSH
  queryShapeHash: 'DE58F0EA1F9D0F0BEC392841E964F3E0A1D246F6636711090F560508D0DE2625',
  command: {
    find: 'book_metadata',
    filter: {
      'reviews.comment': BSONRegExp('Digital', 'i')
    },
    '$db': '2520080053_Ruhitha'
  },
  serverInfo: {
    host: 'Ruhitha',
    port: 27017,
    version: '8.3.8',
    gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
  },
  serverParameters: {
    internalQueryFacetBufferSizeBytes: 104857600,
    internalDocumentSourceGroupMaxMemoryBytes: 104857600,
    internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
    internalQueryFacetMaxOutputDocSizeBytes: 104857600,
    internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
    internalQueryProhibitBlockingMergeOnMongoS: 0,
    internalQueryMaxAddToSetBytes: 104857600,
    internalQueryFrameworkControl: 'trySbeRestricted',
    internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
  },
  ok: 1
}

```

e) Supporting index for common queries (multikey index)

```

>_MONGOSH
> db.book_metadata.createIndex({ formats: 1 });
db.book_metadata.createIndex({ avgRating: -1 });
db.book_metadata.find({ avgRating: { $gt: 4 } }).explain("executionStats");
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: '2520080053_Ruhitha.book_metadata',
    parsedQuery: {
      avgRating: {
        '$gt': 4
      }
    },
    indexFilterSet: false,
    queryHash: 'E21FCB34',
    planCacheShapeHash: 'E21FCB34',
    planCacheKey: '6A4442CC',
    optimizationTimeMillis: 1,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      nss: '2520080053_Ruhitha.book_metadata',
      inputStage: {
        stage: 'IXSCAN',
        nss: '2520080053_Ruhitha.book_metadata',
        keyPattern: {

```

```

>_MONGOSH
      avgRating: -1
    },
    indexName: 'avgRating_-1',
    isMultiKey: false,
    multiKeyPaths: {
      avgRating: []
    },
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'forward',
    indexBounds: {
      avgRating: [
        '[inf, 4)'
      ]
    }
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 45,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,

```

```

>_MONGOSH
  stage: 'FETCH',
  nReturned: 3,
  executionTimeMillisEstimate: 43,
  works: 4,
  advanced: 3,
  needTime: 0,
  needYield: 0,
  saveState: 1,
  restoreState: 1,
  isEOF: 1,
  nss: '2520080053_Ruhitha.book_metadata',
  docsExamined: 3,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 3,
    executionTimeMillisEstimate: 43,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 1,
    restoreState: 1,
    isEOF: 1,
    nss: '2520080053_Ruhitha.book_metadata',
    keyPattern: {
      avgRating: -1
    },
    indexName: 'avgRating_-1',
  },
}

```

```

>_MONGOSH
  find: 'book_metadata',
  filter: {
    avgRating: {
      '$gt': 4
    }
  },
  '$db': '2520080053_Ruhitha'
},
serverInfo: {
  host: 'Ruhitha',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}

```